

Aharonov–Bohm interference in even-denominator fractional quantum Hall states

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Position exchange of non-Abelian anyons affects the quantum state of their system in a topologically protected way¹. Their expected manifestations in even-denominator fractional quantum Hall (FQH) systems offer the opportunity to directly study their unique statistical properties in interference experiments². Here we present the observation of coherent Aharonov–Bohm interference at two even-denominator states in high-mobility bilayer-graphene-based van der Waals (vdW) heterostructures by using the Fabry–Pérot interferometry technique. Operating the interferometer at a constant filling factor, we observe an oscillation period corresponding to two flux quanta inside the interference loop, $\Delta\Phi = 2\Phi_0$, at which the interference does not carry signatures of non-Abelian statistics. The absence of the expected periodicity of $\Delta\Phi = 4\Phi_0$ may indicate that the interfering quasiparticles carry the charge $e^* = \frac{1}{2}e$ or that interference of $e^* = \frac{1}{4}e$ quasiparticles is thermally smeared. Notably, at two hole-conjugate states, we also observe oscillation periods of half the expected value, indicating interference of $e^* = \frac{2}{3}e$ quasiparticles instead of $e^* = \frac{1}{3}e$. To investigate statistical phase contributions, we operated the Fabry–Pérot interferometer (FPI) with controlled deviations of the filling factor, thereby introducing fractional quasiparticles inside the interference loop. The resulting changes to the interference patterns at both half-filled states indicate that the extra bulk quasiparticles carry the fundamental charge $e^* = \frac{1}{4}e$, as expected for non-Abelian anyons.

For more than four decades, quasiparticles carrying fractional charge and obeying fractional statistics have captivated the condensed matter physics community¹. Their most prevalent types are the Abelian anyons, which exhibit quantized exchange phases lying between those of bosons and fermions. Even more notable are non-Abelian anyons that can fundamentally transform the many-body wavefunction through particle exchanges, processing quantum information in a topologically protected manner². The FQH systems have emerged as a leading platform for realizing and manipulating these exotic quasiparticles, owing to high electron mobility, long coherence times and exceptional controllability³. Fractional charge was first observed by means of shot-noise measurements at odd-denominator filling factors expected to host Abelian states^{4,5} and later at the even-denominator filling $\nu = \frac{5}{2}$ in GaAs (refs. 6,7), a leading candidate for non-Abelian topological order⁸.

Direct measurements of anyonic exchange statistics require phase-sensitive techniques such as quantum Hall interferometry in the Aharonov–Bohm regime, in which Coulomb interactions are sufficiently weak for the interferometer area to remain constant as B is varied⁹. Seminal works by Nakamura et al. demonstrated Aharonov–Bohm interference of fractionally charged quasiparticles using a GaAs FPI at filling $\nu = \frac{1}{3}$ (ref. 10), and braiding (double-exchange) phases in

a subsequent study¹¹. These findings were generalized to different filling factors¹² and also observed in alternative platforms, first in integer^{13–15} and then in fractional fillings^{16–18}, and interferometer architectures^{19,20}. In parallel, time-domain braiding experiments^{21–26} also support anyonic quasiparticle statistics in Abelian FQH states²⁷. At even-denominator fillings, Fabry–Pérot interferometry studies at $\nu = \frac{5}{2}$ in GaAs have reported signatures consistent with non-Abelian statistics²⁸. However, the interpretation of those experiments remains challenging, primarily because of the absence of robust Aharonov–Bohm interference.

Even-denominator states have been observed in several FQH platforms, including GaAs (ref. 29), ZnO (ref. 30), graphene³¹, bilayer graphene^{32–35} and WSe₂ (ref. 36). In GaAs narrow quantum wells, thermal-transport measurements^{37,38} consistently support a non-Abelian topological order known as PH-Pfaffian³⁹. Distinct non-Abelian orders known as Moore–Read Pfaffian⁴⁰ and anti-Pfaffian⁴¹ are indicated by daughter states⁴² in bilayer graphene^{35,43} and GaAs wide quantum wells⁴⁴. Specifically, bilayer graphene realizes quantized plateaus at seven half-integer filling factors in the zeroth Landau level. Moreover, the presumed topological orders alternate between Pfaffian and anti-Pfaffian, offering a rich playground for interference studies of non-Abelian anyons.

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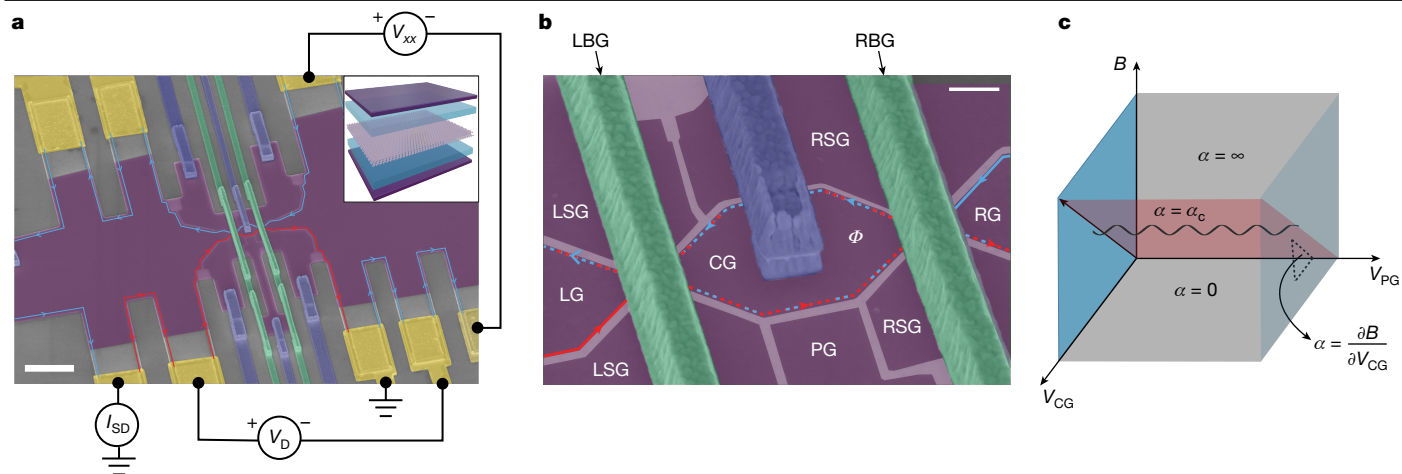


Fig. 1 | FPI based on the bilayer graphene. **a**, Measurement configurations depicted as a false-colour scanning electron microscopy image of the FPI in a bilayer graphene heterostructure (see inset). The top graphite layer (purple) is divided into eight separate regions by means of etched trenches, better seen in **b**. Each region acts as a gate, electrostatically tuned by means of air bridges (blue) to define the interferometer. The device dimensions are indicated by the white scale bar of length 3 μm . In the quantum Hall regime, current I_{SD} applied through an ohmic contact (yellow) propagates by means of edge modes and is

partitioned by two QPCs formed by the left and right split gates (LSG and RSG), resulting in oscillating diagonal resistance $R_D = (V_D^+ - V_D^-)/I_{\text{SD}}$. **b**, Magnification of the interfering region near the centre gate (CG). Left and right air bridges (LBG and RBG, shown in green) are suspended 200 nm above each QPC region, fine-tuning the saddle-point potential. The lithographic interference area, determined by CG area, is 1 μm^2 . Scale bar, 0.3 μm . **c**, Measurement phase space defined by B , V_{PG} and V_{CG} . R_D is measured along planes defined by $\alpha = \frac{\partial B}{\partial V_{\text{CG}}}$ and shown as 2D $B|_{\alpha} - V_{\text{PG}}$ pajamas.

In this work, we report the observation of robust Aharonov–Bohm oscillations at two even-denominator FQH plateaus in bilayer graphene. Using a gate-defined FPI in a high-mobility bilayer-graphene-based vdW heterostructure, we perform a detailed study of the interference patterns as a function of magnetic field, area and density. At both fillings, we observe the unexpected Aharonov–Bohm periodicity $\Delta\Phi = 2\Phi_0$ when the magnetic field and the density are varied together to maintain contact filling. The most conservative interpretation of these measurements is the interference of quasiparticles with charge $e^* = \frac{1}{2}e$, twice the charge expected theoretically⁴⁰ and observed in earlier shot-noise and single-electron transistor measurements in GaAs (refs. 6,7). However, this frequency could also originate from $e^* = \frac{1}{4}e$ quasiparticles performing an even number of loops.

This finding prompted us to study the nearby odd-denominator states at Landau-level fillings of $\nu = \frac{1}{3}$ and $\frac{2}{3}$, for which we found Aharonov–Bohm periodicities corresponding to interference of quasiparticles with charges $e^* = \frac{1}{3}e$ and $\frac{2}{3}e$, respectively. Across the three fillings, the interfering charge follows $e^* = \nu e$ instead of the minimal charges of bulk quasiparticles, which are $\frac{1}{3}e$, $\frac{1}{4}e$ and $\frac{1}{3}e$ for these states. All even-denominator and odd-denominator states for this study are indicated in the phase diagram of bilayer graphene (see Supplementary Information Section 1). We note that, in GaAs, shot-noise measurements at hole-conjugate states also find a partitioned charge of νe (ref. 45) but interference at $\nu = \frac{2}{3}$ shows $e^* = e$ (ref. 10). Finally, by tuning the electron density independently of the magnetic field, we deviate from the fixed-filling constraint, thereby introducing localized bulk quasiparticles⁴⁶. Unlike the integer case, we observe a statistical contribution to the interference phase of fractional fillings, supporting their anyonic character.

Design and measurement phase-space of an FPI

The FPI device is constructed on a high-mobility vdW heterostructure, with bilayer graphene as the active 2D layer, which is encapsulated between hexagonal boron nitride (hBN) dielectric layers, whereas conductive graphite layers on the top and bottom serve as gates. The heterostructure design and nanofabrication techniques follow those detailed in our previous study¹⁶, with measurements conducted under

a perpendicular magnetic field up to $B = 12$ T and at a base temperature of $T = 10$ mK.

A false-colour scanning electron microscopy image of the FPI is shown in Fig. 1a. The top graphite layer is divided into eight distinct regions by 40-nm-wide etched trenches, with each region contacted by means of air bridges. Together with a global graphite back gate, these eight top gates enable capacitive tuning of the potential and displacement fields across various regions of the bilayer graphene. The filling factor inside the interferometer is controlled by the centre gate, whereas the left and right gates set the outer fillings. The two quantum point contacts (QPCs) are formed by the left and right split gates, which set the filling underneath to zero, thereby guiding the counter-propagating edge modes on opposite sides into close proximity and introducing tunnelling between them. An extra plunger gate (denoted PG) allows fine control over the area enclosed by the interfering quantum Hall edge mode. Figure 1b provides a zoomed-in view of the interfering region, lithographically defined to be 1 μm^2 (see Supplementary Information Section 2). Two air bridges, denoted LBG and RBG, positioned 200 nm above the QPC regions, act as gates, fine-tuning the transmission t_r , t_l of each QPC independently. In the measurements presented in this study, $t_{r,l}$ are set to 0.5–0.7 for the interference at the integer states and 0.6–0.9 for the interference at the fractional states (see Supplementary Information Section 3).

We inject a bias current I_{SD} , which propagates along the FQH edge modes with an anticlockwise (clockwise) chirality for electron (hole) carriers, impinging on the FPI as illustrated in Fig. 1a. Current is collected on the other side of the interferometer by a single ground while measuring the diagonal resistance $R_D = \frac{V_D^+ - V_D^-}{I_{\text{SD}}}$ to reveal interference. In the low backscattering regime, R_D includes an oscillatory contribution $\Delta R_D \propto \cos\theta$, in which the interference phase θ is composed of both Aharonov–Bohm (denoted ‘AB’) and statistical phases, that is^{9,47,48},

$$\theta = \theta_{\text{AB}} + \theta_{\text{stat}} = 2\pi \frac{e^*}{e} \frac{AB}{\Phi_0} + N_{\text{qp}} \theta_{\text{anyon}}, \quad (1)$$

in which A is the interfering area, N_{qp} is the integer number of localized quasiparticles within the interference loop and θ_{anyon} is the braiding

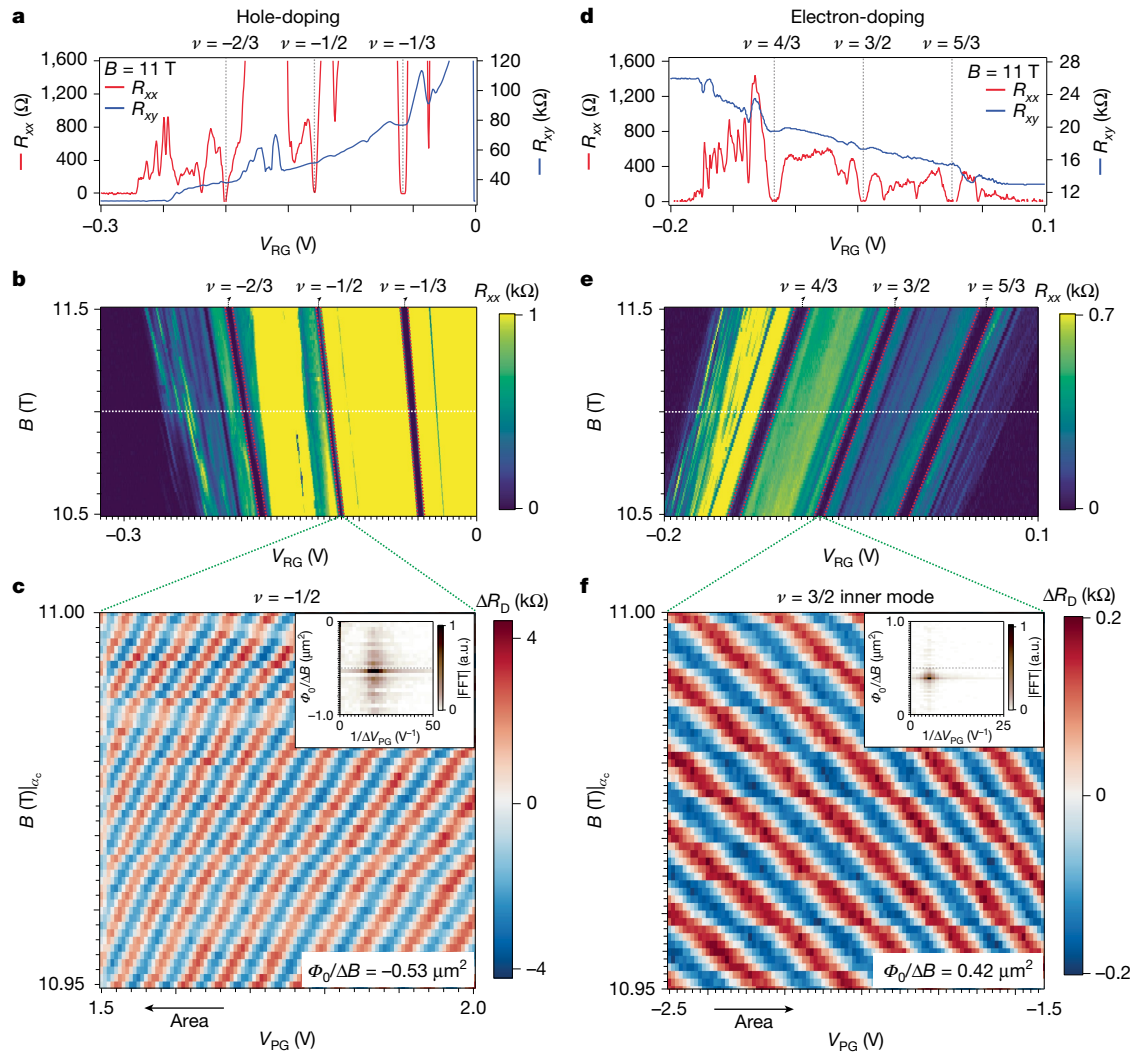


Fig. 2 | Even-denominator Aharonov-Bohm interference. **a**, Longitudinal resistance R_{xx} and Hall resistance R_{xy} measured at 11 T on the right side of the FPI, clearly showing fully developed even-denominator and odd-denominator quantum Hall states at $\nu = -\frac{2}{3}, -\frac{1}{2}$ and $-\frac{1}{3}$. **b**, R_{xx} fan diagram performed on the right side of the FPI between 10.5 and 11.5 T. Dashed red lines indicate the boundaries for each quantum Hall state. **c**, ΔR_D at $\nu = -\frac{1}{2}$ shown as a $B|_{\alpha_c} - V_{PG}$

pajama plot, showing clear Aharonov-Bohm oscillations. Inset, 2D-FFT analysis used to extract the magnetic-field periodicity $\frac{\Phi_0}{\Delta B}$ shown on the lower-right side of the pajama. **d, e**, Same as **a, b** for the electron-doped filling factors $\nu = \frac{4}{3}, \frac{3}{2}$ and $\frac{5}{3}$. **f**, Same as **c** for $\nu = \frac{3}{2}$ with partitioning of the fractional inner mode. a.u., arbitrary units.

phase. For non-Abelian quasiparticles, R_D is predicted to follow a more intricate pattern that differs for even and odd N_{qp} (refs. 49,50).

We perform measurements of R_D in the 3D parameter space spanned by the magnetic field B , the plunger gate voltage V_{PG} and the centre gate voltage V_{CG} ; see Fig. 1c. To disentangle the two terms in θ of equation (1), we follow lines of different slopes $\alpha = \frac{\partial B}{\partial V_{CG}}$ in the $B - V_{CG}$ plane. The Aharonov-Bohm contribution is isolated at the critical trajectories α_c , for which charges are continuously added to the interference loop to maintain constant fillings. Along $\alpha \neq \alpha_c$ trajectories, the filling factor deviates from the rational value defining the quantum Hall plateau through the Hall conductance. All such trajectories are measured within the incompressible region, as shown in Supplementary Information Section 4.

Consequently, R_D follows the well-known ‘pajama pattern’ in the $B - V_{PG}$ plane with a flux periodicity set by the interfering quasiparticle charge e^* . Deviations from this trajectory introduce bulk quasiparticles, N_{qp} , which are expected to manifest individually through phase slips and which alter the average flux periodicity. Other notable trajectories include constant density, $\alpha = \infty$, and constant magnetic field, $\alpha = 0$, illustrated in Fig. 1c.

Even-denominator Aharonov-Bohm interference

We begin the Fabry-Pérot interferometry study of even-denominator states at the filling factor $\nu = -\frac{1}{2}$ owing to its simple edge structure, which consists only of fractional modes. Figure 2a shows the longitudinal resistance R_{xx} and Hall resistance R_{xy} measured at 11 T on the right side of the FPI. The data clearly reveal fully developed integer and fractional quantum Hall states at $\nu = -1, -\frac{2}{3}, -\frac{1}{2}$ and $-\frac{1}{3}$. Figure 2b presents an R_{xx} fan diagram, which we use to extract the constant-filling-factor trajectories. We define $\alpha_c = \frac{\Phi_0}{ve} C$, with $C = \frac{1}{A} \frac{dQ}{dV_{RG}}$ the capacitance per unit area between the right gate and the bilayer graphene underneath, extracted from the Streda formula for each fractional state as the centre of the incompressible region, whose boundaries are indicated by dashed red lines (see Supplementary Information Section 5).

Focusing on the $-\frac{1}{2}$ state, Fig. 2c shows the interference pattern as a function of V_{PG} and $B|_{\alpha_c}$, in which the α_c constraint indicates that V_{CG} is adjusted to maintain constant filling. Specifically, we present the data as $\Delta R_D = R_D - \langle R_D \rangle$, subtracting the average value at each magnetic field. The positive slope of the pajama indicates Aharonov-Bohm-dominated interference, because increasing V_{PG} decreases the interference

area for hole-doped states. The measured visibility, defined as $\text{Visibility} = (G_{\max} - G_{\min}) / (G_{\max} + G_{\min} - 2G_{\text{outer}})$, in which $G_{\max}$ and $G_{\min}$ are the maximum and minimum diagonal conductance values, respectively, and G_{outer} represents the conductance of any fully transmitted outer edge modes, is around 1.9%, comparable with that at integers and odd-denominator states (see Supplementary Information Section 6), and the corresponding edge mode velocity of $v_{\text{edge}} = 7.95 \times 10^3 \text{ m s}^{-1}$, approximately an order of magnitude smaller than that observed at integer filling, is extracted from source-drain bias V_{SD} -dependent R_D (see Supplementary Information Section 7). To extract the flux periodicity, we perform a 2D fast Fourier transform (2D-FFT), shown in the inset of Fig. 2c as a function of $\frac{\phi_0}{\Delta B}$ and $\frac{1}{\Delta V_{\text{PG}}}$. From the magnetic-field periodicity, we extract $A_{\Delta\phi}^{\phi_0} \approx -0.53 \mu\text{m}^2$. The lithographic area $A \approx 1 \mu\text{m}^2$ agrees with that extracted from interference at $\nu = -1$ to within 2% (see Supplementary Information Section 8). Using the same area at $\nu = -\frac{1}{2}$ yields the unexpected flux periodicity $\Delta\phi = (1.89 \pm 0.26)\phi_0 \approx 2\phi_0$. We found that this unexpected $2\phi_0$ flux periodicity was robust against changes in the compressibility of the bulk, magnetic field and in plunger-gate spectroscopy¹³ (see Supplementary Information Sections 10, 11 and 12). Furthermore, our transmission study (Supplementary Information Section 9) shows qualitatively similar interference patterns over the experimentally available $t_{\text{R,L}}$ range. At very low t , at which we expect electron-dominated tunnelling, the visibility is lost.

As shown in Supplementary Fig. 3, nearly all QPCs show resonances, probably because of tunnelling by means of localized states near the saddle-point potential. Such resonances can affect the leading tunnelling channel and the total inference phase⁵¹. Nevertheless, irrespective of the precise tunnelling mechanism across the QPC, the magnetic-field periodicity of the Aharonov–Bohm oscillations at a constant filling provides a direct and robust measure of the fractional charges that dominate tunnelling and interference processes. As long as the QPC environment is approximately constant, owing to low cross-capacitance with the plunger gate and small magnetic-field variations, the total variations in the interference phase owing to Aharonov–Bohm effect and bulk quasiparticles remain unaffected.

Following the first term in equation (1), this periodicity suggests an interfering quasiparticle charge of $e^* = \frac{1}{2}e$, which tunnels across the QPCs to form an interference loop. Quasiparticles with this charge exist as bulk excitations at half-filling, arising from the fusion of two fundamental quasiparticles carrying charge $\frac{1}{4}e$, in all Abelian or non-Abelian FQH candidate states. Alternatively, this periodicity could also arise in a scenario in which non-Abelian $\frac{1}{4}e$ quasiparticles interfere. In that case, when a non-zero number of non-Abelian quasiparticles are localized in the bulk, there are several degenerate ground states. Fluctuations between these ground states on the timescale of the measurement could suppress the $4\phi_0$ periodicity that arises from a single winding of $\frac{1}{4}e$ quasiparticles while not affecting the $2\phi_0$ periodicity arising from double windings or $\frac{1}{2}e$ quasiparticles^{49,50}.

To test the generality of these findings, we investigated the $\nu = \frac{3}{2}$ plateau (on the electron side), which exhibits a gap comparable with $\nu = -\frac{1}{2}$ (refs. 33,34). Similar to the previous case, Fig. 2d shows R_{xx} and R_{xy} measured at 11 T, revealing well-developed FQH states at $\nu = \frac{4}{3}, \frac{3}{2}$ and $\frac{5}{3}$. Figure 2e presents an R_{xx} fan diagram, which we use to extract α_c as before. Figure 2f shows the interference pattern as a function of V_{PG} and $B|_{\alpha_c}$ that arises when the QPCs are tuned to partition the fractional inner edge mode (see Supplementary Information Section 13 for the interference of the integer outer edge). The slope of the pajama pattern with 5.6% visibility is opposite to $\nu = -\frac{1}{2}$, indicating Aharonov–Bohm-dominated interference for electron-doped states (see Supplementary Information Section 14). The 2D-FFT, shown in the inset of Fig. 2f, yields the magnetic-field periodicity $A_{\Delta\phi}^{\phi_0} \approx 0.42 \mu\text{m}^2$. Estimating the interfering area based on the integers $\nu = 1$ and 2, we find $A = 0.99 \pm 0.10 \mu\text{m}^2$, consistent with the lithographic area (see Supplementary Information Section 15). Using this area at $\nu = \frac{3}{2}$,

we conclude $\Delta\phi = (2.35 \pm 0.78)\phi_0 \approx 2\phi_0$. Temperature-dependence measurements showed a reduction in visibility with increasing temperature, whereas the magnetic-field periodicity remained constant (see Supplementary Information Sections 16 and 17).

These two measurements consistently show periodicities close to $2\phi_0$ and not the expected $4\phi_0$. The observations reflect the interference of $\frac{1}{2}e$ quasiparticles at $\nu = -\frac{1}{2}$ and $\frac{3}{2}$. We note that the topological orders of electrons at both fillings are believed to be Pfaffians^{35,43} but their edge structures at the boundary to $\nu = 0$ are qualitatively different. In particular, a Pfaffian order of electrons at $\nu = -\frac{1}{2}$ is equivalent to an anti-Pfaffian of holes. Consequently, it exhibits an edge state with three upstream Majorana fermions at a $\nu = 0$ boundary. Insofar as the identification of these states is accurate, our experiment effectively examines two distinct non-Abelian topological orders.

Comparing fractional quasiparticle interference

At both even-denominator filling factors, the observed Aharonov–Bohm periodicity is consistent with an interfering quasiparticle charge that matches the Landau-level filling factor $\nu_{\text{LL}} = \frac{1}{2}$. In this study, ν_{LL} is the filling factor of the partially filled Landau level $\nu_{\text{LL}} = \nu - \lfloor \nu \rfloor$. The interfering charge at $\nu = \frac{1}{3}$ also follows the filling factor^{10,17,18}. We extended the study to states at $\nu = -\frac{2}{3}$ and $\frac{5}{3}$ to determine whether their interfering charge is also set by the filling or by the minimal bulk excitation. We refer to these states as ‘hole-conjugate’ based on their presumed topological orders and edge structures, which are related to the ‘particle-like’ states at $\nu = -\frac{1}{3}$ and $\frac{4}{3}$ by a hole conjugation in the partially occupied Landau level. As there is no particle–hole symmetry, the microscopic wavefunctions at these fillings are not related by hole conjugation.

Figure 3a,b shows the extracted flux periodicities for all six fractional fillings in constant-filling measurements. The values for the odd denominators are extracted from the pajama patterns in Fig. 3c–f through the 2D-FFTs shown in Fig. 3g–j, assuming the same interference areas for hole-doped and electron-doped states as before. The results confirm the interference of $e^* = \nu_{\text{LL}}e$ quasiparticles in all states included in our study. A recent experiment on the hole-conjugate states $\nu = \frac{2}{3}, \frac{5}{3}$ and $\frac{4}{3}$ in GaAs using a Mach–Zehnder interferometer also observed interference of $e^* = \nu_{\text{LL}}e$ quasiparticles²⁰. It is not understood why non-fundamental quasiparticles should dominate the interference, as our measurements at half-filled and hole-conjugate states indicate. We point out that previous interference experiments at the hole-conjugate $\nu = \frac{2}{3}$ state in GaAs reported the periodicity $\Delta\phi = \phi_0$ corresponding to the interference of electrons¹⁰. Moreover, Mach–Zehnder interference of the higher particle-like Jain states $\nu = \frac{2}{5}$ and $\frac{3}{7}$ observed $\Delta\phi = 5\phi_0$ and $7\phi_0$ (ref. 19), corresponding to the fundamental quasiparticle charge instead of $\nu_{\text{LL}}e$.

Theoretically, the question of which type of quasiparticle tunnel is addressed on the basis of the renormalization of bare tunnelling amplitudes by the interactions intrinsic to fractional edge modes. The bare tunnelling amplitudes for different quasiparticles are non-universal and hard to calculate reliably. Their renormalization, encoded by means of a scaling dimension of tunnelling operators, is the same for fundamental and $e^* = \frac{2}{3}e$ quasiparticles at the $\nu = \frac{2}{3}$ edge⁵². It is possible that interactions across the QPC tip the balance in favour of $e^* = \frac{2}{3}e$ tunnelling. Alternatively, when both tunnelling processes occur with comparable probabilities, the $3\phi_0$ periodicity expected for $\frac{1}{3}e$ quasiparticles could be thermally suppressed because it requires exciting a neutral mode, which propagates with a much smaller velocity than the charge mode. At half-filling, the scaling dimensions of $e^* = \frac{1}{4}e$ quasiparticles depend on which topological state is realized but their numerical values are generally close to those of $e^* = \frac{1}{2}e$ tunnelling. Different interactions across the QPC could favour either and a similar thermal suppression may affect the $\frac{1}{4}e$ quasiparticle, which also excites a neutral mode. We also warn that the

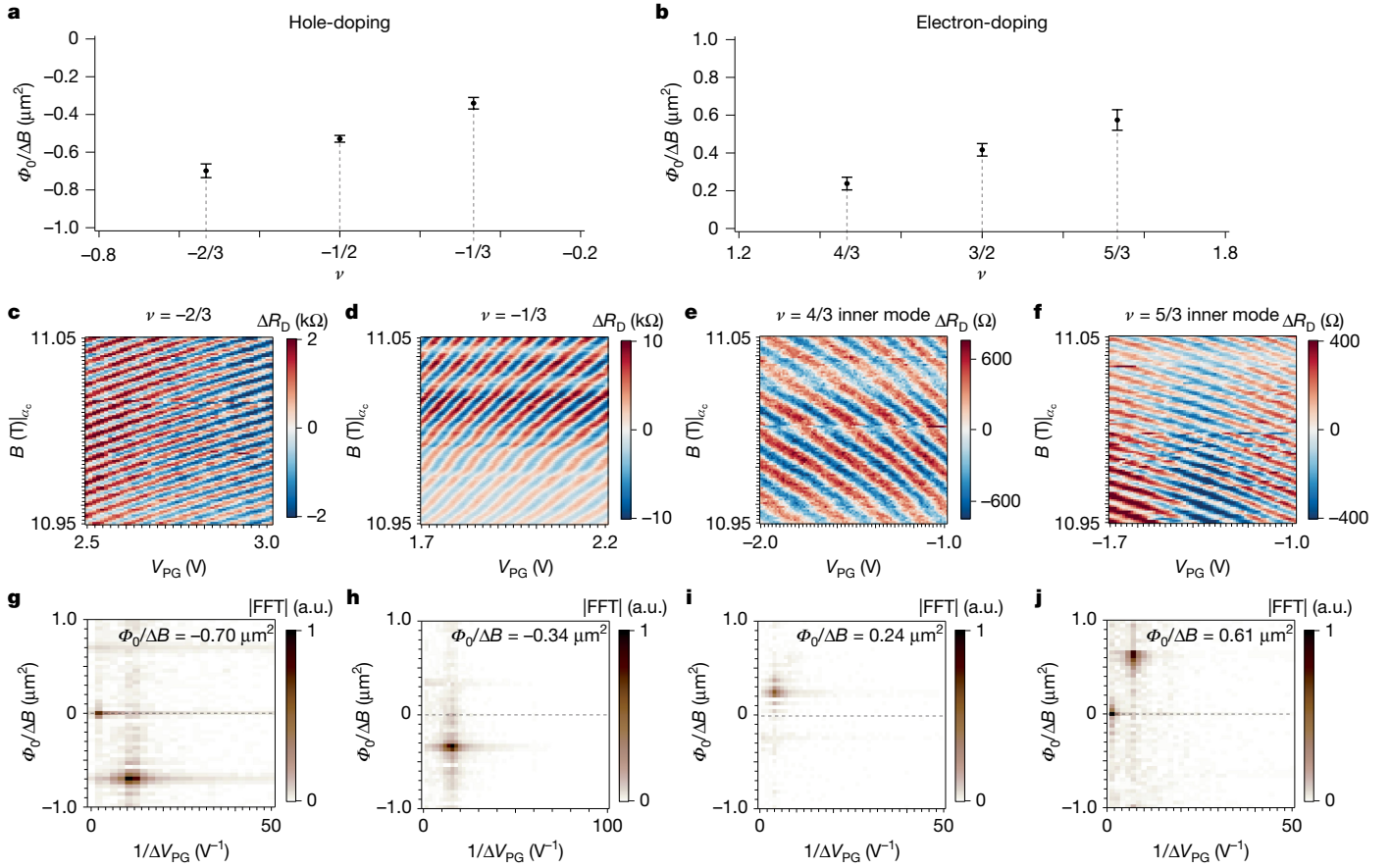


Fig. 3 | Interference of $e^* = \nu_{LL}e$ quasiparticles in various FQH states.

a, Magnetic-field periodicities $\phi_0/\Delta B$ at constant filling extracted from 2D-FFT analyses at $\nu = -\frac{2}{3}, -\frac{1}{2}$ and $-\frac{1}{3}$. Error bars correspond to the variance of the Gaussian fit obtained by fitting the 2D-FFT with a Gaussian function (see Supplementary Information Section 8). **b**, Same for $\nu = \frac{4}{3}, \frac{3}{2}$ and $\frac{5}{3}$ with partitioning of the fractional inner mode. Error bars correspond to the variance of the Gaussian fit obtained by fitting the 2D-FFT with a Gaussian function (see Supplementary Information

Section 15). **c–f**, ΔR_D shown as $V_{PG} - B|_{\alpha_c}$ pajamas for $\nu = -\frac{2}{3}, -\frac{4}{3}, \frac{4}{3}$ and $\frac{5}{3}$, respectively. At $\nu = \frac{5}{3}$, α deviated from α_c by 3%. **g–j**, Corresponding 2D-FFTs, along with the extracted magnetic-field periodicities. We show all 2D pajama plots measured with partitioning of all of the fractional modes, before subtracting the average value at each magnetic field to discuss how changes in B and V_{CG} affect the QPC transmission (see Supplementary Information Section 18). a.u., arbitrary units.

scaling dimensions extracted from experiments often deviate substantially from theoretical expectations.

Statistical phase from bulk anyons in FPI

Interference of fractional quasiparticles fundamentally differs from that of electrons by quantum statistical effects, that is, the second term in equation (1). Interfering quasiparticles acquire a quantized phase change for each localized anyon in the interferometer bulk. For electron interference, this extra phase is an unobservable multiple of 2π independent of the bulk anyon type. To observe such contributions, we operate the FPI at $\alpha \neq \alpha_c$, such that tuning the magnetic field or V_{CG} causes the filling factor to deviate slightly from the rational value ($\nu = \frac{p}{q}$) reflected by the bulk Hall conductivity. These deviations introduce excess charge carriers in the form of quasiparticles inside the interference loop. Each well-isolated quasiparticle in the bulk is expected to result in a sharp phase jump in the interference pattern. Similar to previous studies^{11,16,18,19}, we observe such discrete phase slips at $\nu = -\frac{1}{3}$, which are analysed in Supplementary Information Sections 19 and 20. Introducing quasiparticles at a constant rate along a fixed α trajectory alters the overall slope of the constant-phase lines in the pajama pattern. This change in slope is more robust against fluctuations in the number of bulk quasiparticles around its mean than individual phase slips. Such fluctuations can broaden and even entirely wash out phase slips (see Supplementary Information Section 21).

We attribute the absence of phase slips at both half-fillings and hole-conjugate states to such fluctuations and proceed with a careful analysis of the slope.

The change of the slope provides crucial insights into which quasiparticles enter the interference loop as the filling factor varies. In Supplementary Information Section 21, we show theoretically that the slope can even distinguish between different bulk quasiparticle types being introduced as ν changes. For the case in which interfering quasiparticles entering equation (1) carry charge $e^* = \nu_{LL}e$ and fundamental quasiparticles are introduced into the bulk, we find

$$\begin{aligned} \text{Integer edge modes: } \quad \frac{\phi_0}{\Delta B} &= A = \text{constant}, \\ \text{Fractional edge modes: } \quad \frac{\phi_0}{\Delta B} &= (\nu_{LL} - \nu)A + \nu \frac{\alpha_c}{\alpha} A. \end{aligned} \quad (2)$$

Because the mutual statistics with the $e^* = \frac{1}{2}e$ quasiparticles with all other quasiparticles is Abelian, equation (2) holds for all paired states. The second line matches the phase $\theta = 2\pi\langle N \rangle$, with N the number of electrons in the loop, generalizing the result in ref. 46. If filling-factor deviations introduce quasiparticles other than the fundamental ones into the bulk, the slope on the right-hand side ($\nu\alpha_c A$) changes. We warn that strong bulk–edge coupling would also affect the slope and lead to an α dependence of the Aharonov–Bohm pattern even at integer fillings. We analyse this scenario in detail in Supplementary Information Sections 22 and 23 and show that it does not apply to our

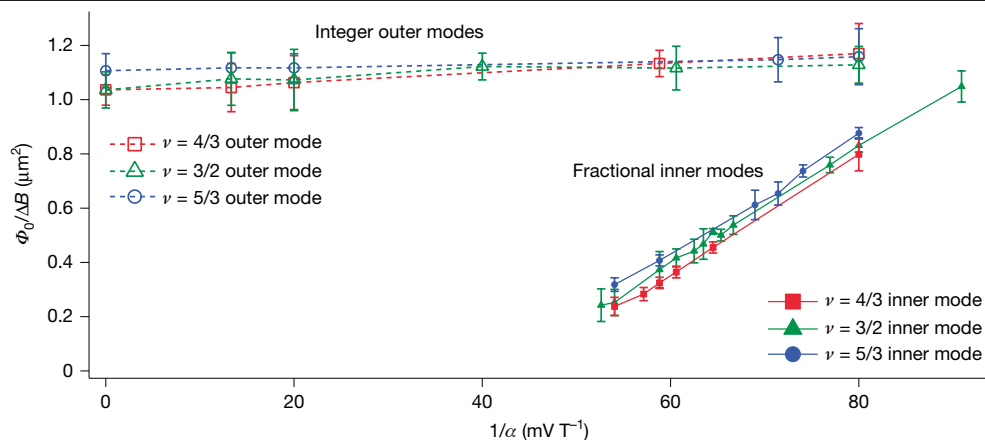


Fig. 4 | Statistical contribution to the interference of fractional quasiparticles. Magnetic-field periodicities $\frac{\Phi_0}{\Delta B}$ obtained along different trajectories α from 2D-FFTs for the fractional inner and integer outer modes

observations in the fractional case (see Supplementary Information Section 24).

We extract $\frac{\Phi_0}{\Delta B}$ from the 2D-FFT for the fillings $\nu = \frac{4}{3}, \frac{3}{2}$ and $\frac{5}{3}$, for partitioning of both the fractional inner modes and the integer outer modes. Figure 4 shows our results for each α , with $\frac{\Phi_0}{\Delta B}$ obtained from the 2D-FFT of the corresponding pajama patterns. In Supplementary Information Section 25, we show all 2D pajama images and corresponding 2D-FFT analyses for all fractional filling factors discussed in this study. For all of the integer outer modes, ΔB is independent of the α , as expected. By contrast, for the fractional modes, all measurements collapse into a single linear dependence on $\frac{1}{\alpha}$ as in the second line of equation (2). Their slope deviates by 15% from the numerical value expected on the basis of the bulk capacitance C , obtained through the Streda formula for the region to the right of the FPI. This discrepancy can arise from boundary effects of the comparatively small centre gate, small changes in the interference area with V_{CG} and bulk–edge couplings (see Supplementary Information Section 26). At $\nu = \frac{3}{2}$ with an interfering inner mode, the observed slope matches the theoretically expected statistical contribution of bulk $e^* = \frac{1}{4}e$ quasiparticles to the interference of Abelian $e^* = \frac{1}{2}e$ quasiparticles.

Conclusions

Our results mark two notable advancements towards the long-standing goal of observing non-Abelian statistics, bringing this objective within reach. The first essential condition—quasiparticle coherence in candidate non-Abelian states—is demonstrated by the observation of Aharonov–Bohm interference at two even-denominator FQH states. The second condition—interference contributions from localized bulk quasiparticles—was also observed by tuning the magnetic field and density to deviate from constant filling. Notably, our observations indicate that the localized bulk quasiparticles exhibit a charge of $\frac{1}{4}e$, as expected for non-Abelian quasiparticles.

The observed flux periodicity of $2\Phi_0$ is consistent with the non-Abelian double-winding scenario or with a scenario in which it arises from the interference of Abelian $\frac{1}{2}e$ quasiparticles. Distinguishing between these two possibilities is crucial for conclusively identifying non-Abelian behaviour and could be achieved through shot-noise measurements at a single QPC. If the Abelian $\frac{1}{2}e$ quasiparticles are indeed responsible for the interference signal, developing techniques to facilitate the tunnelling of fundamental quasiparticles at the QPC will be essential. Adjusting the saddle-point potential or screening of the interedge interactions could affect the characteristics of the QPCs in this way. Our observation of an apparent interfering charge being twice the fundamental one at $\nu = \frac{2}{3}$ and $\frac{5}{3}$ indicates that Abelian

of $\nu = \frac{4}{3}, \frac{3}{2}$ and $\frac{5}{3}$. Error bars correspond to the variance of the Gaussian fit obtained by fitting the 2D-FFT with a Gaussian function (see Supplementary Information Sections 15, 25).

hole-conjugate states can provide valuable insights on how to control tunnelling of different quasiparticle types. Resolving this question in bilayer graphene could permit direct observation of non-Abelian statistics at several distinct FQH phases.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09891-2>.

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Stack preparation

In this study, we use vdW heterostructures in which a bilayer graphene layer is encapsulated between hBN and graphite layers. To prepare flakes from bulk graphite crystals, SiO₂/Si substrates are cut into 10-mm × 10-mm pieces. These pieces are placed on tape, which is used to exfoliate the bulk crystals. The exfoliated flakes are then transferred onto the SiO₂/Si substrate and heated on a hotplate at 170–180 °C for 90 s. When the tape has cooled, these pieces are removed to search for the desired bilayer graphene and graphite flakes. After cooling, the tape is removed and the flakes are inspected to identify suitable bilayer graphene and graphite layers. hBN flakes are prepared in a similar manner by exfoliating bulk hBN crystals using thin polydimethylsiloxane (PDMS). The stack is prepared using polycarbonate stamps held with Kapton tape, placed on a diamond-shaped PDMS layer on top of a glass slide. To ensure strong adhesion between the polycarbonate film and the PDMS, the stamps are placed on a hotplate at 170–180 °C for 2 h. The transfer stage is heated to 130–131 °C, allowing the sequential pickup of all vdW layers in the following order: top graphite, top hBN, bilayer graphene, bottom hBN and bottom graphite. A thickness of 29 (27) nm for the top (bottom) hBN and 5 nm for the top and bottom graphite is used in the fabricated device. The estimated bulk-gate capacitance C_b is approximately 2.2×10^{-15} F, yielding the bulk charging energy $E_c = 36$ μ eV. The prepared stack is transferred at 180 °C on a clean SiO₂/Si substrate and left for roughly 15 min at 180 °C to melt the polycarbonate and detach it from the PDMS. The polycarbonate film is dissolved by placing the sample in chloroform for 3–4 h and subsequently cleaned with isopropyl alcohol and deionized water. Subsequently, the stack on the SiO₂/Si substrate undergoes thermal annealing in an ultrahigh vacuum (about 10^{−9} torr) at 400 °C for 4 h and 30 min to remove residual contaminants and bubbles. Last, we use atomic force microscopy (AFM) ironing, for which contact-mode AFM is used with 100-nN mild force applied on the stack, to flatten the stack on an atomic scale by suppressing the random strain fluctuations induced while stacking. This ironing with AFM can improve the quality of the bulk, resulting in an increase in the gap of quantum Hall states.

Device fabrication

The bilayer-graphene-based electronic FPI is fabricated on a five-layer vdW heterostructure placed on a highly p-doped Si substrate with a 280-nm SiO₂ oxide layer, using standard nanofabrication and lithography techniques. The process begins with creating alignment markers for the electron-beam lithography and bonding pads, using Ti (10 nm)/Au (60 nm)/Pd (20 nm). The device geometry is defined through reactive ion etching with poly(methyl methacrylate) (PMMA) resist serving as the etch mask. To etch out two main materials used in the heterostructure, hBN and few-layer graphite layers, O₂/CHF₃ mixture with the volume ratio of 1:10, and O₂ are used as etching gas for hBN and graphite, respectively. After defining the geometry, the sample undergoes thermal annealing in ultrahigh vacuum (about 10^{−9} torr) at 350 °C for 2 h and 30 min for resist residual removal on the stack. A trench approximately 40 nm wide is etched into the top graphite using mild O₂ plasma conditions to minimize damage to the top hBN layer, dividing the top graphite into eight sections. Finally, bridges are fabricated to independently tune the potential of each graphite section. This is achieved using PMMA/MMA/PMMA trilayer resists, followed by a 20-s mild O₂ plasma etch and the subsequent evaporation of Cr (5 nm)/Au (320 nm). To achieve highly transparent contacts between the metal and the bilayer graphene, the edge contacts are fabricated in the last step by etching the top hBN layer with O₂/CHF₃, followed by angled evaporation of Cr (2 nm)/Pd (20 nm)/Au (60 nm), avoiding that the contacts are exposed to lots of heat during device fabrication.

Measurements

The device is measured in a highly filtered dilution refrigerator at the base temperature of 10 mK using the standard low-frequency lock-in amplifier technique. An SRS 865A lock-in amplifier generates an alternating voltage at 13.7 Hz and measures the voltage difference between two contacts. A 100-M Ω load resistor is included in series with the lock-in amplifier, allowing the system to source an alternating current ranging from 50 to 0.5 nA. A QDAC, an ultralow-noise 24-channel digital-to-analogue converter (Qdevil-QM), is used to tune the voltages applied to all graphite gates and the two air bridges. Furthermore, a Keithley 2400 voltage source is used to apply a voltage to the highly p-doped Si substrate, doping the contact region and improving the contact resistance.

Plunger-gate spectroscopy

Plunger-gate spectroscopy³³ (Supplementary Figs. 6 and 15) reveals a divergence of the interference frequency for integer and most of the particle-like fractional states as the edge goes beneath the gate, consistent with a rapidly increasing gate–edge capacitance and enabling extraction of the edge–gate distance^{13,14}. For hole-conjugate and even-denominator states, the oscillation visibility is suppressed as the edge approaches the gate, precluding direct resolution of the divergence within the accessible range. We attribute this contrast to a more complex edge structure and greater sensitivity to local disorder near the etched trenches. From the plunger-gate dispersion of the interference frequency (Supplementary Fig. 13), we extract the gate–edge separation. For integer and particle-like fractional states, the inferred distance decreases from about 70 nm and about 150 nm, respectively, to nearly zero as the edge moves under the plunger gate. These values are consistent with previous integer-regime measurements^{13,14} and extend the approach to odd-denominator fractional fillings (see Supplementary Information Section 12). For hole-conjugate and even-denominator states, the oscillations lose visibility as the edge nears the gate, obscuring the expected frequency divergence and preventing a reliable distance extraction over the accessible range. In these cases, QPC characterization (Supplementary Fig. 3f–i) identifies the participating edge by means of step-like transmission with plateaus, consistent with a fully transmitted integer outer mode.

Data availability

The data supporting the plots in this paper, as well as all custom analysis code written in Igor for data fitting and visualization, are available from the online depository Zenodo at <https://doi.org/10.5281/zenodo.17454013> (ref. 54).

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R.K. developed the measurement circuit and a dilution refrigerator. J.K. performed the measurements. J.K., H.D., A. Shaer, A. Stern, D.F.M. and Y.R. analysed the measured data. A. Shaer, A. Stern and D.F.M. developed the theoretical aspect. J.K., H.D., A. Shaer, A. Stern, D.F.M. and Y.R. wrote the paper, with input from all authors. Y.R. supervised the overall work carried out on the project.

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Additional information

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